
Online Shoppers Purchase Prediction System

Project Id: AIML-21

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1. Problem Description

The objective of this project is to analyze online shopping session data and prepare the dataset for predicting whether a user session will result in revenue (a purchase) or not. E-commerce platforms generate large amounts of user interaction data such as number of pages visited, time spent on pages, bounce rates, exit rates, visitor type, and session timing. Analyzing this data helps businesses understand customer behavior, improve targeted marketing, and increase sales by identifying potential buyers.

The dataset used in this project contains both numerical and categorical features related to user browsing behavior and a target variable named Revenue, which indicates whether a purchase was made. In this phase of the project, the dataset was cleaned and preprocessed by removing duplicate records, checking for missing values, identifying numerical and categorical features, and analyzing data distributions. Exploratory Data Analysis (EDA) was performed using various visualization techniques such as distribution plots, box plots, count plots, and correlation heatmaps to understand feature relationships and patterns in the data.

Feature engineering techniques such as log transformation, feature scaling, categorical encoding, and preprocessing pipeline creation using ColumnTransformer and Pipeline were performed to prepare the dataset for machine learning models. The processed data will be used in the next phase for model training and evaluation.

2. Architecture

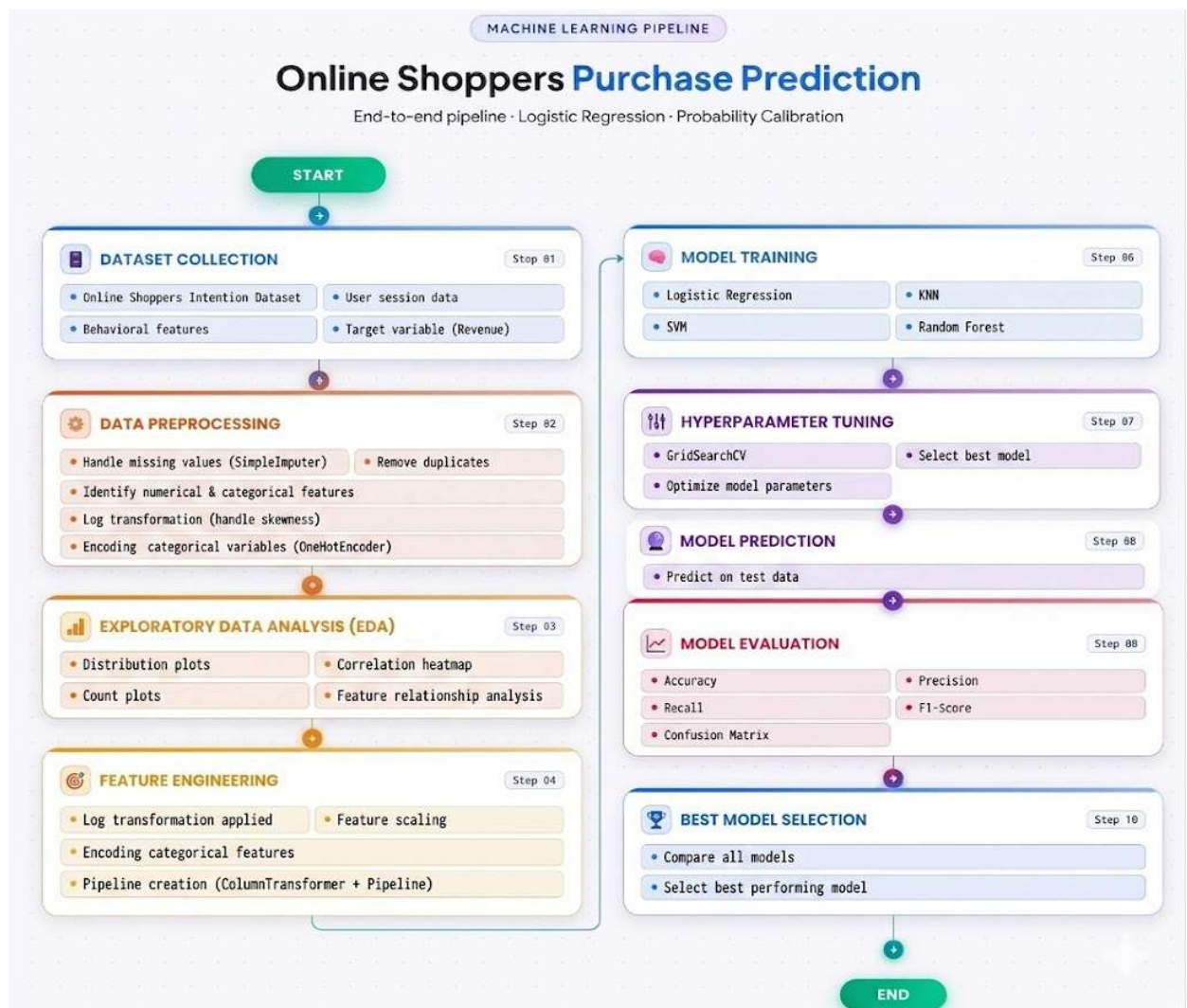


Fig.1 Sample flow diagram.

3. Models Used(Applied/ Planned) Give a brief introduction of the models used

In this phase, multiple machine learning models were implemented and evaluated to identify the best-performing model for predicting online shopper purchase behavior. The models were trained on preprocessed data and compared using evaluation metrics such as accuracy, precision, recall, and F1-score. Hyperparameter tuning was applied to improve model performance.

(i) *Logistic Regression*

Logistic Regression is a linear classification algorithm used to predict binary outcomes. It applies a sigmoid function to convert feature combinations into probability values between 0 and 1.

4. Used as a baseline model
5. Handles binary classification effectively
6. `class_weight='balanced'` was used to address class imbalance
7. Hyperparameters (such as C, penalty, solver) were tuned using GridSearchCV

(ii) *K-Nearest Neighbors (KNN)*

KNN is a distance-based algorithm that classifies a data point based on the majority class of its nearest neighbors.

- Simple and intuitive model
- Sensitive to feature scaling (handled using StandardScaler)
- Hyperparameter tuned: number of neighbors (k)
- Works well for pattern-based classification but may struggle with large datasets

(iii) *Support Vector Machine (SVM)*

SVM is a powerful classification algorithm that finds the optimal hyperplane to separate different classes.

- Effective in high-dimensional feature spaces
- Can handle both linear and non-linear classification using kernels
- Hyperparameters tuned: C (regularization), kernel type, gamma
- Provides strong performance but is computationally expensive

(iv) *Random Forest*

Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

- Handles non-linear relationships well
- Reduces variance by averaging multiple trees

- Hyperparameters tuned: number of estimators, max depth
- Performs well but may reduce interpretability

4. Evaluation metrics used

In this phase, different machine learning models were evaluated using standard classification metrics. These metrics help measure how well the models predict whether a user will make a purchase (Revenue = 1) or not (Revenue = 0).

(i) *Accuracy*

Accuracy measures the overall correctness of the model by calculating the proportion of correct predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Indicates how many predictions are correct out of total predictions
- May be misleading in imbalanced datasets

(ii) *Precision*

Precision measures how many of the predicted positive cases (buyers) are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

- High precision means fewer false positives
- Important when false alarms (predicting buyer incorrectly) should be minimized

(iii) *Recall*

Recall measures how many actual positive cases (buyers) are correctly identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

- High recall means fewer missed buyers
- Important in this project since identifying potential buyers is critical

(iv) F1 Score

F1 Score is the harmonic mean of precision and recall, providing a balance between both.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

- Useful when there is class imbalance
- Gives a balanced measure of model performance

(v) Confusion Matrix

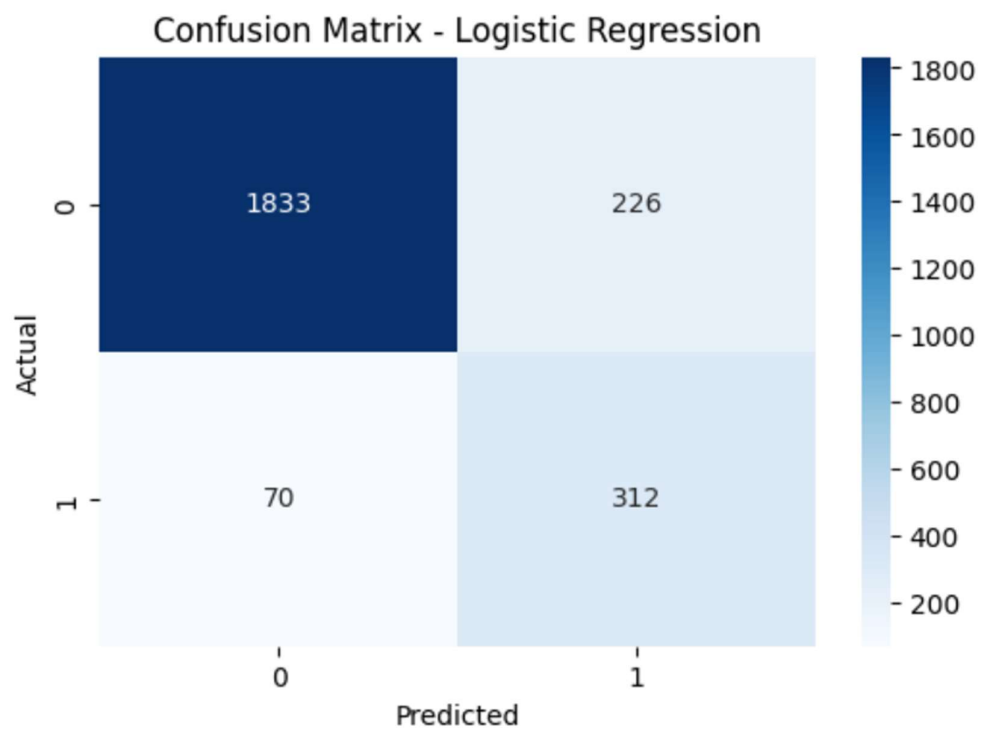
Confusion Matrix provides a detailed breakdown of prediction results by comparing actual and predicted values.

$$\begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix}$$

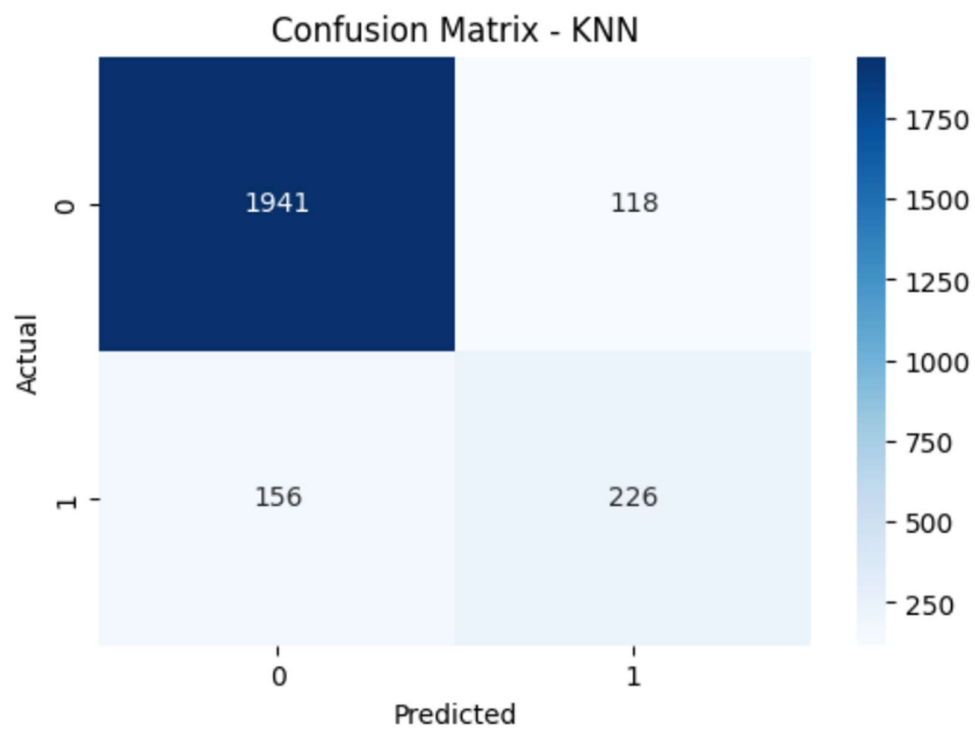
Where:

- TP (True Positive) – Correctly predicted buyers
- TN (True Negative) – Correctly predicted non-buyers
- FP (False Positive) – Incorrectly predicted buyers
- FN (False Negative) – Missed buyers

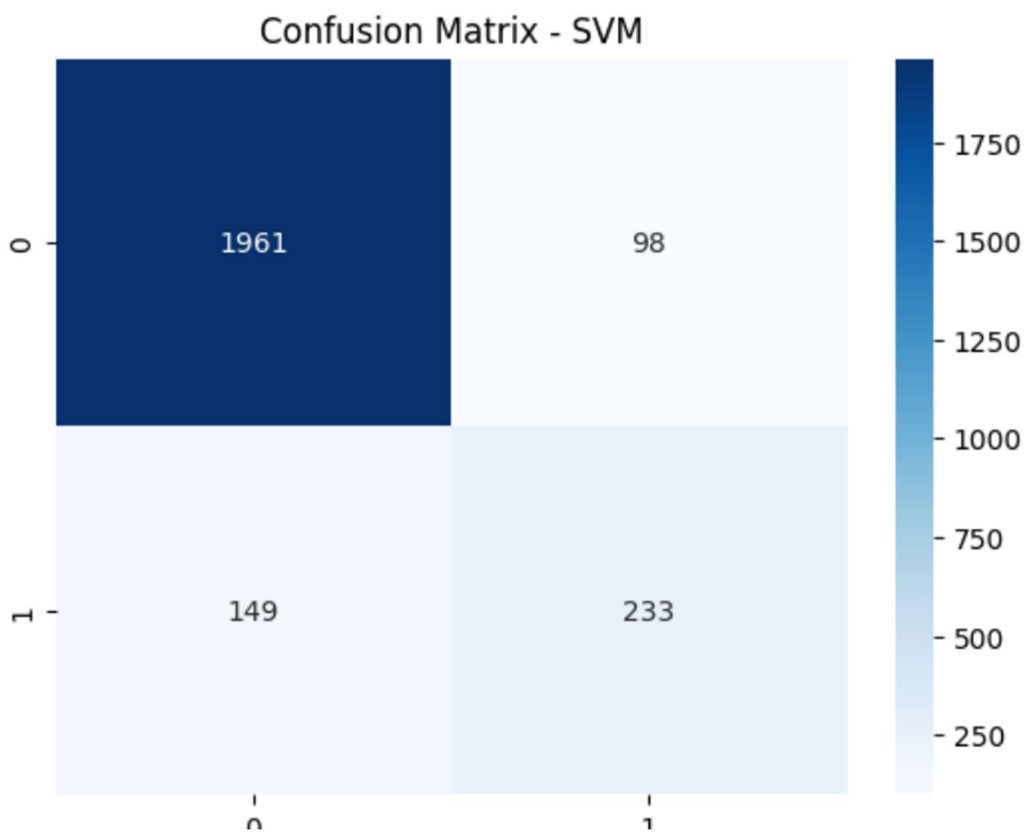
5. Preliminary Results and Visualizations



Cross-validation scores: [0.93732077 0.88529291 0.84145842 0.83490373 0.81114297]
Mean CV accuracy: 0.8620237607537893

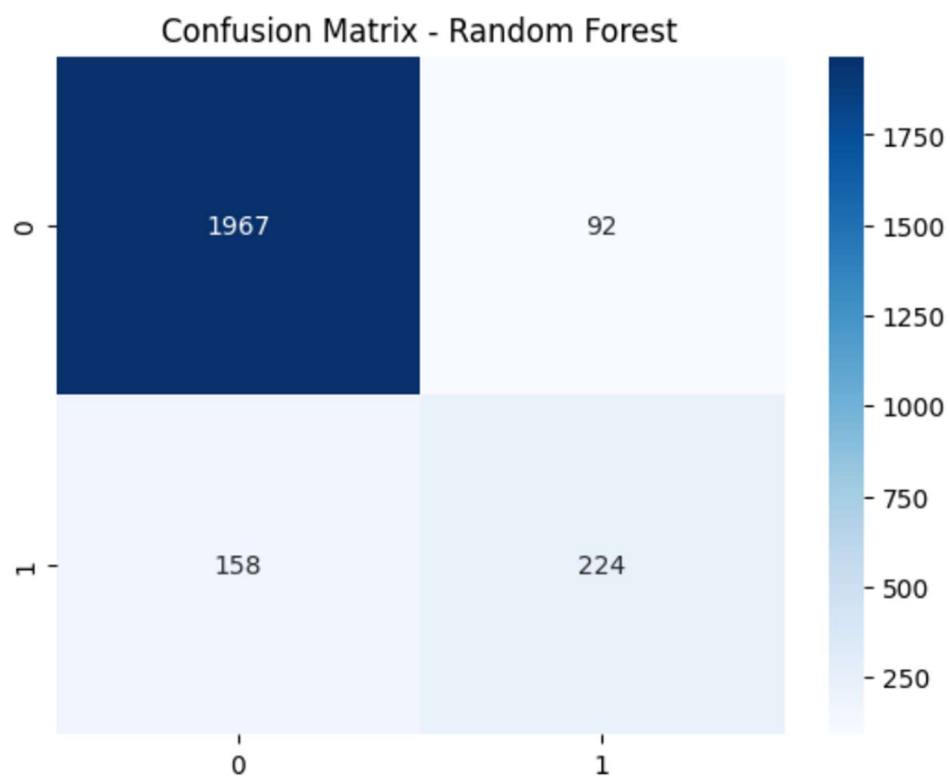


	precision	recall	f1-score	support
False	0.93	0.95	0.94	2059
True	0.70	0.59	0.64	382
accuracy			0.90	2441
macro avg	0.81	0.77	0.79	2441
weighted avg	0.89	0.90	0.89	2441



Classification Report:

	precision	recall	f1-score	support
False	0.93	0.95	0.94	2059
True	0.70	0.61	0.65	382
accuracy			0.90	2441
macro avg	0.82	0.78	0.80	2441
weighted avg	0.89	0.90	0.90	2441



	precision	recall	f1-score	support
False	0.92	0.95	0.94	2059
True	0.70	0.57	0.63	382
accuracy			0.89	2441
macro avg	0.81	0.76	0.78	2441
weighted avg	0.89	0.89	0.89	2441

6.References:

Books:

1. Let Us Python – Yashavant Kanetkar
2. [Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow](#) (3rd Edition, 2022)

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- TutorialsPoint – <https://www.tutorialspoint.com/cprogramming/index.htm>
- Stack Overflow – <https://stackoverflow.com>
- ClaudeCode - <https://claude.com>